

To find the following Machine Learning Regression method using in r2 value:

1. Multiple Linear Regression (**R2_Score**) = **0.9358**

2. SUPPORT VECTOR MACHINE:

S.No.	HYPER PARAMETER	LINEAR (r value)	RBF (NON LINEAR) (r value)	POLY (r value)	SIGMOID (r value)
1	C=0.1	0.9375	-0.0575	-0.0568	-0.0575
2	C=1.0	0.8951	-0.0573	-0.0509	-0.0575
3	C=10	-2.4372	-2.4372	0.0253	-0.0576
4	C=100	-357.08	-0.0302	0.4657	-0.0588
5	C=500	Not predicted	0.0500	0.6208	-0.0640
6	C=1000	Not predicted	0.1606	0.6403	-0.0707
7	C=2000	Not predicted	0.2884	0.6717	-0.0845
8	C=5000	Not predicted	0.5035	0.7663	-0.1297
9	C=10000	Not predicted	0.6743	0.8142	-0.2174
10	C=20000	Not predicted	0.7930	0.7549	-0.5653

The SVM Regression use R2 value (**Linear and Hyper Parameter (C=0.1)**) = **0.9375**

3. DECISION TREE:

S.No.	CRITERION	MAX FEATURES	SPLITTER	R_VALUE
1	Mse	auto	best	0.92479
2	Mse	auto	random	0.86926
3	Mse	sqrt	best	0.71506
4	Mse	sqrt	random	0.81364
5	Mse	log2	best	0.87138
6	Mse	log2	random	0.69858
7	Mae	auto	best	0.94869
8	Mae	auto	random	0.89351
9	Mae	sqrt	best	0.58774
10	Mae	sqrt	random	0.73293
11	Mae	log2	best	0.95308
12	Mae	log2	random	0.84460

13	Friedman_mse	auto	best	0.92740
14	Friedman_mse	auto	random	0.86646
15	Friedman_mse	sqrt	best	0.52946
16	Friedman_mse	sqrt	random	0.57883
17	Friedman_mse	log2	best	0.40064
18	Friedman_mse	log2	random	0.86844
19	poisson	auto	best	0.92601
20	poisson	auto	random	0.92214
21	poisson	sqrt	best	0.39666
22	poisson	sqrt	random	0.53052
23	poisson	log2	best	0.52743
24	poisson	log2	random	0.64234

The Decision Tree Regression use R2 value (Mae, 'log2', 'best') = **0.95308**